

Image Viewer Integrity Evaluation System

User Manual

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Disclaimer

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1 Introduction

The Image Viewer Integrity Evaluation System (IVIES) for whole slide imaging devices is a Medical Device Development Tool (MDDT) and is categorized as a Non-clinical Assessment Model (NAM). It is a non-clinical, colorimetry-based test method that measures the device performance of pathology whole slide image viewing software devices to determine the need for additional performance study. The MDDT is not an FDA-cleared or approved medical device. The official MDDT summary can be found at <https://www.fda.gov/medical-devices/medical-device-development-tools-mddt>.

1.1 Context of Use

The Image Viewer Integrity Evaluation System (IVIES) is a software tool qualified to assist with assessing whether a pathology whole slide image viewing software device can decode the same input image file and display pixelwise identical images when compared to the predicate whole slide imaging device image review manipulation software. The IVIES tool quantitatively measures the differences between two images by calculating their colorimetric difference for each pixel. The colorimetric pixel differences can be used as a quantitative measure to compare the performance of two whole slide image viewing software devices and to determine whether additional clinical study data is needed to further characterize the performance of the device.

IVIES is not suitable for evaluation of third-party viewers that intentionally alter the image rendering or the downstream components such as the computer environment or the display system.

1.2 Tool Description and Principle of Operation

The IVIES is a software tool that measures the output differences between two pathology whole slide image (WSI) viewing software devices. The IVIES compares the images generated by two devices and reports their pixelwise differences. The IVIES can be used to assist with assessing whether a third-party pathology WSI viewing software device (“third-party viewer”) can decode the same input image file and display pixelwise identical images when compared to the predicate WSI device image review manipulation software (“predicate IRMS”).

The IVIES analyzes image datasets generated by both devices displaying the same WSI files across multiple regions of interest, magnification levels, and tissue specimens. The IVIES performs colorimetric analysis using standard color space conversions (CIEXYZ and CIELAB) and calculates pixelwise color differences with the CIEDE2000 standard. Based on FDA-established tolerance thresholds, the IVIES determines whether images are identical and generates a structured analysis report including dataset table of contents and comparison results.

1.3 Background

The pathology whole slide imaging (WSI) system referred to in this document is defined as *“an automated digital slide creation, viewing, and management system intended as an aid to the pathologist to review and interpret digital images of surgical pathology slides. The system generates digital images that would otherwise be appropriate for manual visualization by conventional light microscopy. The whole slide imaging system is not intended for use with slides prepared from frozen tissue, cytology, and non-FFPE hematopathology specimens (e.g., peripheral blood smear).”* per product code PSY, under 21 CFR §864.3700. As stated in [FDA Guidance - Technical Performance Assessment \(TPA\)](#)

of Digital Pathology Whole Slide Imaging Devices¹, the components in a pathology WSI system are grouped into two subsystems: the *image acquisition* and the *image display*. In 2017, the first pathology WSI system was granted marketing authorization through DEN160056 and comprises the scanner, software, and display². Considering the imaging pipeline, this configuration can be simplified as three cascading components: scanner, viewer, and display, as shown in Figure 1. The *scanner component* digitizes the tissue slide as a digital image file in a specific WSI file format. The *viewer component* decodes the digital image file and drives the *display component* to generate optical stimuli to reproduce the original image for the human user to visualize.

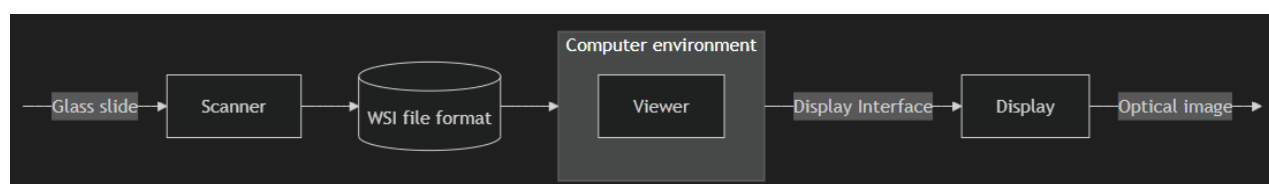


Figure 1: A WSI system comprises three components -- scanner, viewer, and display.

In 2020, Sectra Digital Pathology Module (K193054), a software-only device³ was cleared with the product code QKQ⁴, as a standalone digital pathology image viewing and management software that can be used in lieu of the original image review manipulation software (*IRMS*, as defined in the FDA TPA guidance) in the predicate pathology WSI system for viewing WSI files generated by the intended scanner⁵. In this document, such devices are referred to as “*third-party viewers*” because they are usually manufactured by a third-party vendor and are not included in the original pathology WSI system’s clinical study. As shown in Figure 2, the third-party viewer and the predicate *IRMS* are interchangeable software applications that share the same upstream scanner and file format, as well as the downstream display. The two software applications run in the same computer environment, which includes hardware and software components such as the operating system, browser, color management, color profile, graphics card driver, and anything that may alter the pixels before sending to the display. Therefore, the substantial equivalence between the third-party viewer and the predicate *IRMS* can be established by demonstrating that the digital images generated by both are pixelwise identical in lieu of conducting a clinical study to demonstrate substantial equivalence. To be precise, for the same slide, when the same region of interest (ROI) is viewed at the same magnification level, the third-party viewer and the predicate *IRMS* should generate pixelwise identical digital images. The rationale is that the display component will produce the same optical stimuli for each pixel if the input digital images are pixelwise identical (cf. “Image1” and “Image2” in Figure 2). If the optical images are identical, then the human user will not be able to distinguish the differences. Thus, a clinical study that relies on the human user’s visual interpretation may not be necessary to demonstrate substantial equivalence in a marketing application. Such a test is called a *pixelwise comparison study* (PWC)⁶.

¹ <https://www.fda.gov/media/90791/download>

² https://www.accessdata.fda.gov/cdrh_docs/reviews/DEN160056.pdf

³ https://www.accessdata.fda.gov/cdrh_docs/pdf19/K193054.pdf

⁴ <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpd/classification.cfm?id=5478>

⁵ https://www.accessdata.fda.gov/cdrh_docs/pdf19/K190332.pdf

⁶ https://www.accessdata.fda.gov/cdrh_docs/reviews/K193054.pdf

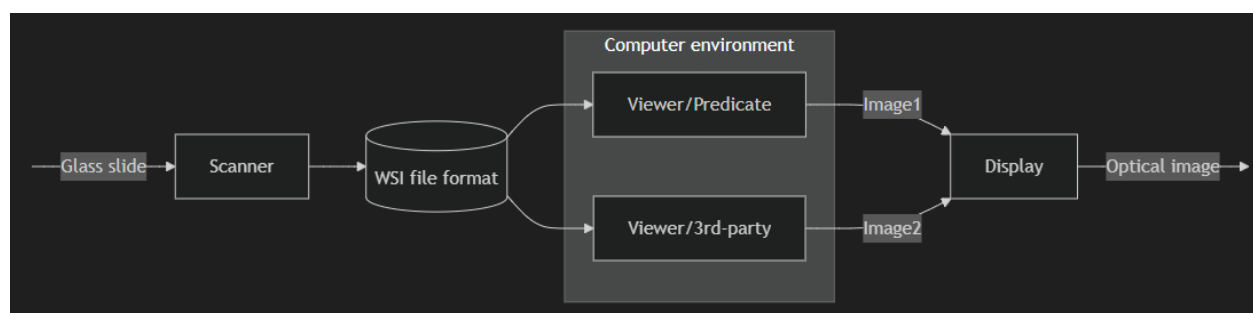


Figure 2: The third-party viewer shares the same upstream and downstream components as the predicate IRMS.

The sponsor should refer to the most recent decision summary for a digital pathology image viewing and management software device (Product cleared under the device product code QKQ) or contact FDA/CDRH/OHT7 for the study design, minimum sample size requirements as well as the ΔE_{00} criteria. General guidelines for conducting a pixelwise comparison study is provided below:

- Test with multiple slides across multiple regions of interest (ROI) at multiple magnification levels.
- Use Hematoxylin and Eosin (H&E) Formalin-Fixed Paraffin-Embedded (FFPE) glass slides of normal and tumor tissues from various human anatomical organs.
- Select ROIs at different locations that include the biological features identified by qualified pathologist or other qualified personnel while minimizing the blank areas.
- Use an operating system-level software tool (e.g., Microsoft Snipping Tool) to capture the images displayed by the viewer on the intended display as screenshots saved in an uncompressed file format such as PNG. The tool should be validated for capturing the actual pixel data sent to the display.
- Ensure the screenshots were cropped and registered to be pixelwise comparable. The cropped image needs to include most of the pixels in the image except for those in the viewer-specific user interface areas.

Figure 3 illustrates the preparation, submission and review process of a typical pixelwise comparison test data. In the sponsor's domain, the sponsor conducts the pixelwise comparison study and compiles the test data following the aforementioned guidelines provided by the FDA. The sponsor submits the test dataset to the FDA as part of the marketing application using the eSTAR program through the [CDRH Portal](https://www.fda.gov/medical-devices/industry-medical-devices/send-and-track-medical-device-premarket-submissions-online-cdrh-portal)⁷. In the FDA domain, the review team then downloads and inspects the dataset. If the submitted dataset is incomplete or inadequate, the review team will request the missing information, and the sponsor may need to rectify and repeat the above process to provide updated test data. Otherwise, the dataset will be analyzed by the review team with the assistance of the MDDT to determine whether the predicate IRMS and the subject viewer produce identical images. The MDDT generates a table of contents (ToC) of the dataset that is shared by both the sponsor and the FDA to facilitate the communication and analysis processes.

When preparing the pixelwise comparison test data, each filename or folder name should follow a consistent naming convention and be programmatically accessible using the indices of slide, ROI, magnification level, and browser, referring to Appendix A. Given the upload

⁷ <https://www.fda.gov/medical-devices/industry-medical-devices/send-and-track-medical-device-premarket-submissions-online-cdrh-portal>

size limit of CDRH Portal per submission, the sponsor may need to submit the image dataset via different amendments. It is highly recommended that the sponsors design a naming convention such that the image datasets can be split into separate subfolders that can be independently uploaded to separate folders or “AMENDMENTS” in the CDRH Portal. The naming convention should establish a suitable folder structure to balance the number of folders and the maximum folder size.

Non-PDF files (e.g., excel files, binary image files) should be added to the ZIP file and placed under the “MISC FILES” folder when uploading the information to the CDRH Portal. More information regarding the technical limitations can be found in [Send and Track Medical Device Premarket Submissions Online: CDRH Portal](#)⁸ and [eCopy Medical Device Submissions](#)⁹.

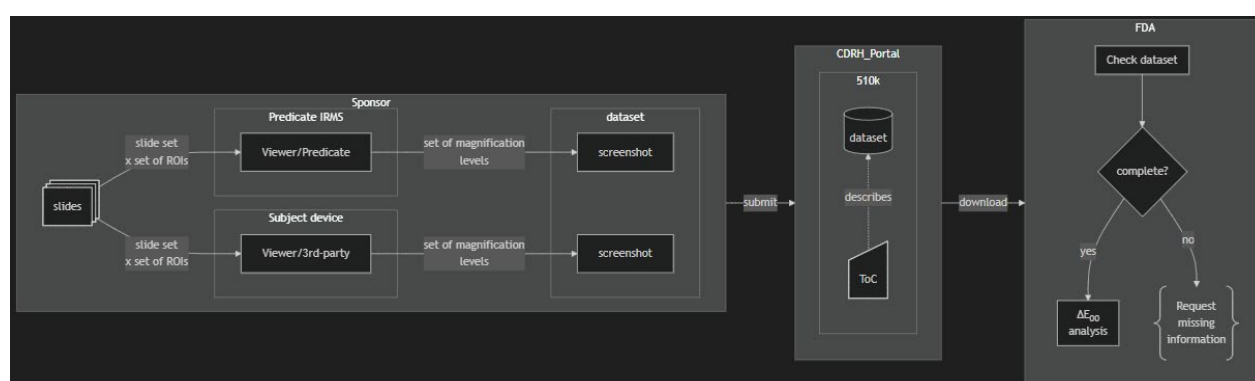


Figure 3: Preparation, submission and review process of the pixelwise comparison test data.

1.4 Tool Principle

Figure 4 illustrates the five steps used in the MDDT to compare two images:

- The MDDT assumes that the predicate IRMS and the third-party viewer are connected to a hypothetical display that is based on the *Standard RGB (sRGB)* color space [1] (although the viewer devices may be intended for operation with a display based on a different color space). Based on the sRGB standard, the MDDT calculates the expected stimuli, generated by the hypothetical display, for each pixel in the CIE XYZ color space.
- The MDDT converts each pixel from the CIE XYZ color space to the CIELAB color space using the CIELAB standard [2] and the white point of the hypothetical display.
- For each corresponding pixel pair between the predicate IRMS and the third-party viewer, the MDDT calculates the color difference, expressed as ΔE_{00} , using the CIEDE2000 standard [3].
- Among all ΔE_{00} values calculated from an image pair, the MDDT calculates the 95th percentile. The purpose is to exclude the highest ΔE_{00} values occurring in odd cases such as the image boundaries.
- Based on the tolerance ΔE_{00} determined by FDA/CDRH/OHT7/DMGP (e.g., equal to or less than $3 \Delta E_{00}$), the MDDT concludes whether the two images are identical.

⁸ <https://www.fda.gov/medical-devices/industry-medical-devices/send-and-track-medical-device-premarket-submissions-online-cdrh-portal>

⁹ <https://www.fda.gov/medical-devices/how-study-and-market-your-device/ecopy-medical-device-submissions>

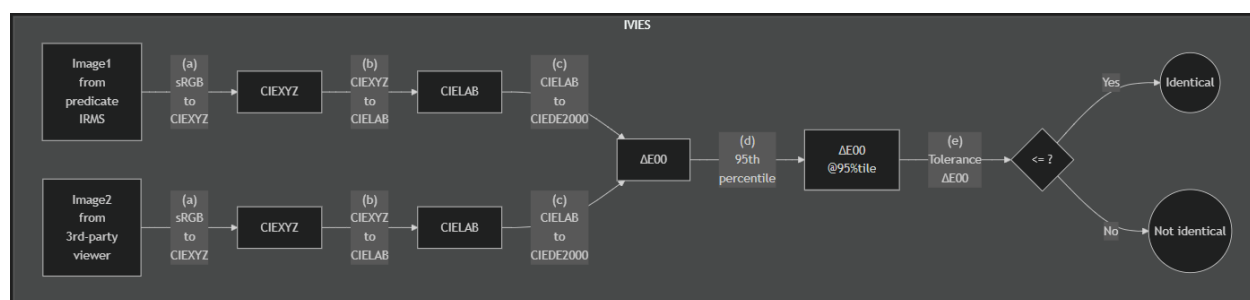


Figure 4: Steps of comparing two images pixelwise.

Figure 5 shows the ΔE_{00} distribution of a sample image-pair, which consists of 1,281,582 pixel-pairs. The left plot shows the distribution of the ΔE_{00} values. Although the distribution has a peak at 0.7 ΔE_{00} , many pixels have a ΔE_{00} greater than 3 (the right tail passing the green line). The right plot shows the normalized cumulative curve of the same distribution. The red line indicates the 95th percentile intercepting the curve at 3.19 ΔE_{00} . If the tolerance is given as ≤ 3 ΔE_{00} , the MDDT will conclude that these two images are not identical.

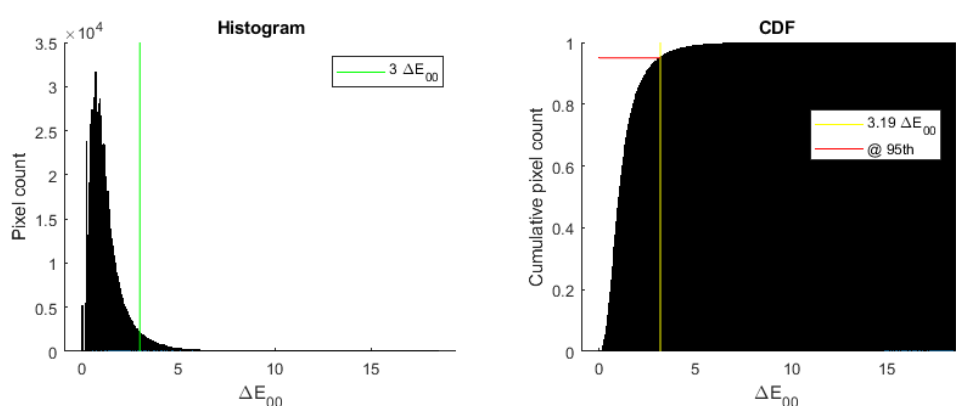


Figure 5: ΔE_{00} distribution of a sample image-pair.

1.5 Method of Measurement

The MDDT is implemented in MATLAB¹⁰. As illustrated in Figure 4, Step (a) sRGB-to-CIE XYZ, Step (b) CIE XYZ-to-CIE LAB, and Step (c) CIE LAB-to-CIEDE2000 are collectively implemented by the following MATLAB function:

```
imcolordiff(image1, image2, "Standard", "CIEDE2000", "isInputLab", false);
```

where

The “image1” and “image2” variables indicate the two input images.

The “isInputLab” parameter is set to “false” so the conversion of “image1” and “image2” from the sRGB color space to the CIE LAB color space should be conducted.

The “Standard” and “CIEDE2000” parameter-pair specifies that the color difference calculation should use the CIEDE2000 formula.

¹⁰ The MathWorks, Inc. MATLAB. Version R2023a, The MathWorks, Inc., 2025.

Step (d) 95th percentile is implemented by the following MATLAB function:

```
prctile(imdiff1,95);
```

where the “imdiff1” variable contains all ΔE_{00} values and the “95” parameter indicates the 95th percentile.

2 Install IVIES

2.1 System Requirements

Component	Requirement
Operating System	Windows 10 or later (64-bit)
MATLAB Runtime	R2023a (9.14a)
Memory (RAM)	8 GB minimum, 16 GB recommended
Storage	Varies based on dataset size; recommend 50 GB free space on a local or network drive
Permissions	Administrator rights for installation of the MATLAB Runtime Permission to execute the downloaded .exe application

2.2 Install MATLAB Runtime

The MATLAB Runtime R2023a (9.14a) library must be installed before running the compiled IVIES executable. The library is freely available for downloading at:

<https://www.mathworks.com/products/compiler/matlab-runtime.html>

Installation Steps:

- Download MATLAB Runtime R2023a (9.14a)
- ① **NOTE:** Only the R2023a (9.14a) version can be used because the MATLAB Runtime version must match the compiled app.
- Run the MATLAB Runtime installer and follow the on-screen instructions
- ① **NOTE:** Installing the MATLAB Runtime does not need a MATLAB license.
- Verify installation at: C:\Program Files\MATLAB\MATLAB Runtime\R2023a

2.3 Install IVIES

Download the IVIES:

- Visit <https://github.com/DIDSR/ivies/releases>
- Download the executable file: IVIES_04162025.exe
- Save the file to a convenient location on the computer
- No installation required - the application runs directly from the executable

3 Prepare the Datasets

As shown in Figure 3, the user needs to prepare image datasets generated by both devices displaying the same WSI files across multiple regions of interest, magnification levels, and tissue specimens. Please see FDA's general guidelines for the pixelwise comparison study (page 8) and the file naming convention in Appendix A.

The IVIES provides a simplified sample dataset for demonstration purposes. In the “demo” folder, there are 3 subfolders:

- “demo\input\predicate” contains 48 test images generated by the predicate viewer (see Figure 6)
- “demo\input\subject” contains 48 test images generated by the subject viewer (see Figure 7)
- “demo\overview” contains 8 WSI overview images showing the ROIs (see Figure 8)

In the following steps, the sample datasets will be used to explain how to provide the required information to IVIES.

3.1 Identify the data location

The user needs to provide the location of the datasets. In this example, the root folder of the sample datasets is “N:\demo”.

3.2 Understand the naming convention

Under the “demo” folder, the “predicate” folder contains 48 test images from the predicate device, and the “subject” folder contains 48 corresponding test images from the subject device. According to the file listing shown in Figure 6 and Figure 7, the file naming convention uses the following format:

```
<s>-<r>-<m>.png
```

where

```
s := 017, 036, 072, 074, 096, 108, 128, 131
r := roi1, roi2, roi3
m := 20x, 40x
```

The “overview” folder contains 8 WSI overview images. According to the file listing shown in Figure 8, the file naming convention uses the following format:

```
<s>.png
```

where

```
s := 017, 036, 072, 074, 096, 108, 128, 131
```

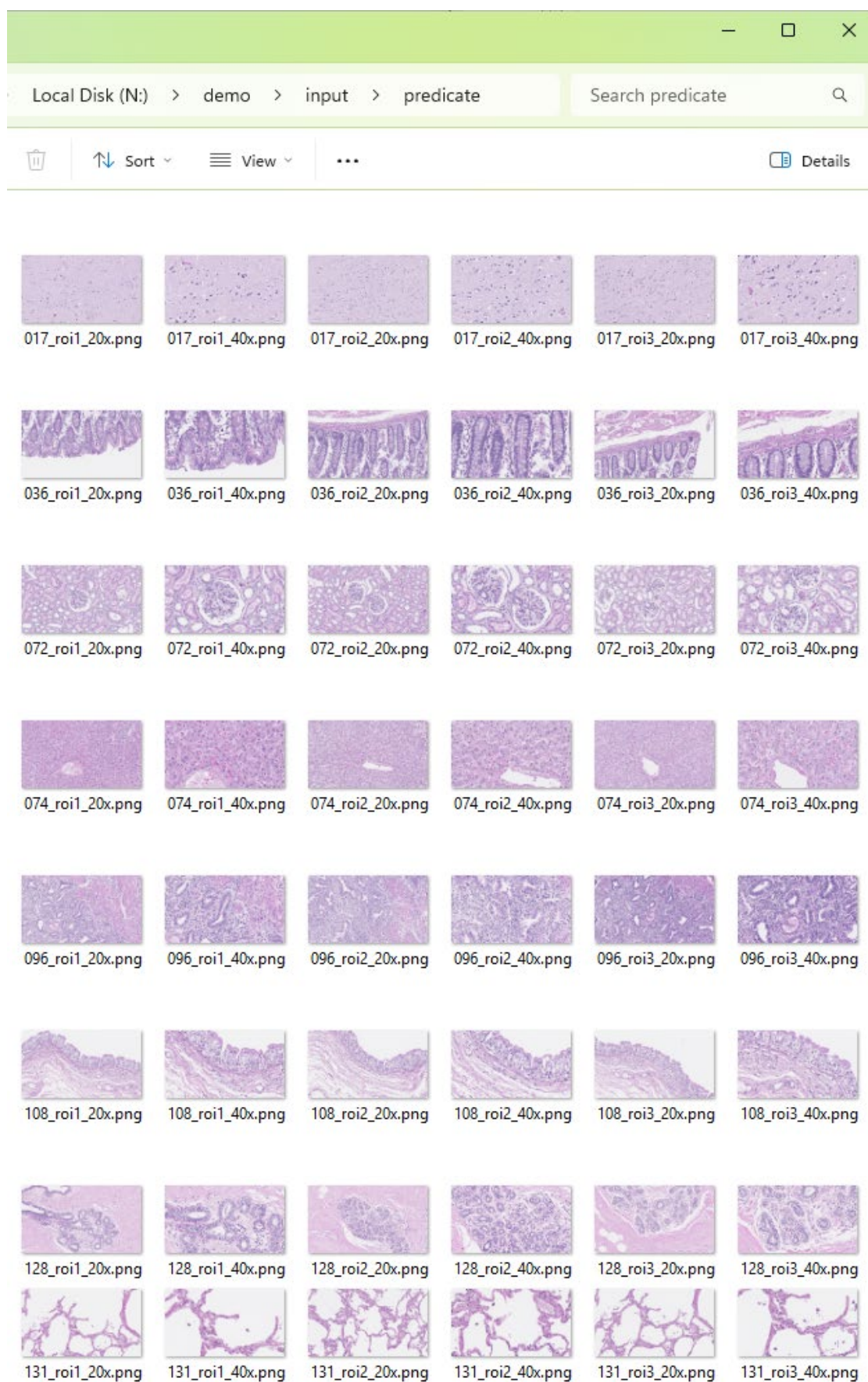


Figure 6: The 48 test images from the predicate device in the “demo” folder.



Figure 7: The 48 test images from the subject device in the “demo” folder.

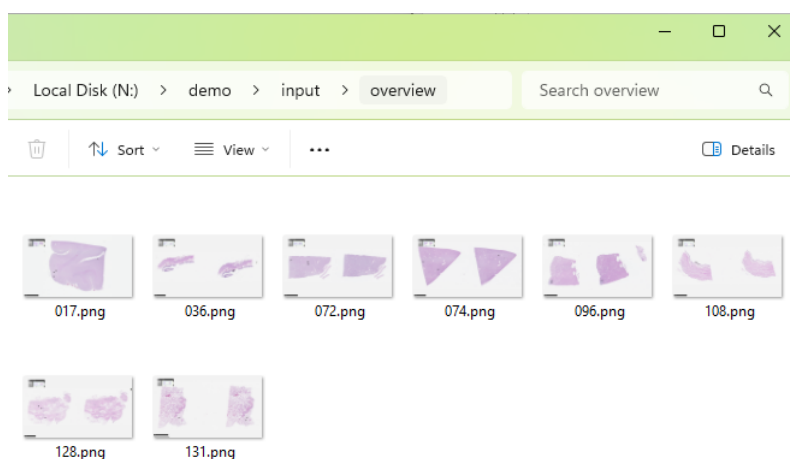


Figure 8: The WSI overview images in the “demo” folder.

4 Use IVIES

To start the IVIES app, double-click on the “IVIES 04162025.exe” icon (Figure 9).

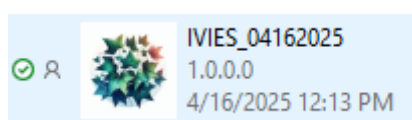


Figure 9: Start the app.

The IVIES graphic user interface (GUI) shows up in a few seconds (Figure 10). The IVIES GUI is an interactive form for the user to provide information that describes the datasets.

Image Viewer Integrity Evaluation System

Submission_Info

K-number*

Sponsor

Device

FileFormat

Root folder

File_Folder_Naming_Convention

Slide# i = 1 : slide_n* Using

Formula_Slide

Gen_Slide

ROI# j = 1 : roi_n*

Formula_ROI

Gen_ROI

MAG# k = 1 : mag_n*

Formula_Mag

Gen_MAG

Device# w = 1 : device_n*

Formula_Device

Gen_Device

Slide_Name (s)

1	017
2	036
3	072
4	074

ROI_Name (r)

1	roi1
2	roi2
3	roi3

MAG_Name (m)

1	20x
2	40x

Device_Name (d)

1	
2	

Folder_Slide_Name (ss)

1	
2	
3	

Folder_ROI_Name (rr)

1	
2	
3	

Folder_MAG_Name (mm)

1	
2	

Folder_Device_Name (dd)

1	predicate
2	subject

RTA_Analysis

Create_RTA

Run_All

Check_Completeness

Gen_ROI_chart

Comp# x = 1 : comp_n*

	Predicate	Subject
1		1
2		2

Calculate_dE

Store_Load_Form

Output_filename

Input_filename

Screenshot_Filename

Folder_Naming*

File_Naming*

Gen_Pathname

	Sl...	ROI#	Ma...	De...	Pathname	Found
1	1	1	1	1	predicate/017_roi1_20x.png	☒
2	1	1	1	1	subject/017_roi1_20x.png	☒
3	1	1	1	2	predicate/017_roi1_40x.png	☒
4	1	1	1	2	subject/017_roi1_40x.png	☒
5	1	2	1	1	predicate/017_roi2_20x.png	☒

WSI_Overview_Filename

WSI_Name_using

WSI_Naming

Check_WSI_name

	WSI_pathname	Found
1	overview/017.png	☒
2	overview/036.png	☒

Generate_Inventory_Report

Generate_dE_Report

☒ Include Completeness chart

☒ Include ROI chart

☒ Include WSI chart

☒ Include dE chart

dE_Tolerance =

OnePager

Text_Report

Figure 10: The initial GUI screen.

The IVIES GUI includes the “tooltip” function that provides a real-time prompt for each user-input field. The user can move the cursor inside an input field to see the instructions (Figure 11).

The screenshot shows the main GUI of the Image Viewer Integrity Evaluation System. A tooltip is displayed over the 'Device' input field in the 'Submission_Info' panel. The tooltip text reads: 'Enter the data location on the local computer or network file system. It usually contains both a subfolder for the test images and a subfolder for the WSI overview images.' The GUI is divided into several panels: Submission_Info, Display_Info, File_Folder_Naming_Convention, Screenshot_Filename, WSI_Overview_Filename, RTA_Analysis, Store_Load_Form, and Generate_dE_Report. Each panel contains various input fields, buttons, and tables for configuring the system.

Figure 11: The tooltip.

The IVIES GUI contains 9 panels as shown in Figure 12. Their functions are described in the following 9 sub-sections. The below section 4 explains the function of each panel.

The screenshot shows the same GUI as Figure 11, but with nine panels numbered 1 through 9. The panels are: 1. Submission_Info, 2. Display_Info, 3. File_Folder_Naming_Convention, 4. Screenshot_Filename, 5. WSI_Overview_Filename, 6. Generate_dE_Report, 7. RTA_Analysis, 8. Store_Load_Form, and 9. Generate_dE_Report. Each panel is highlighted with a red border and a red number. The panels contain various input fields, buttons, and tables for configuring the system.

Figure 12: The nine panels in the IVIES GUI.

4.1 Panel 1: Enter Submission Information

The information entered in panel #1 is used to identify the present study. There are 5 fields for the user to enter (Figure 13).

- K-number: It will be used to suggest filenames for saving/loading data in panel #8. This field is required. In this example, “K345678” is a made-up submission ID.
- Sponsor name: It will be included in the report as the title. This field is optional.
- Device name: It will be used in the report as the title. This field is optional.
- File format, browser, or any other notes such as amendments, supplements, date, etc.: It will be used to suggest filenames for saving/loading data in panel #8. This field is optional.
- Root folder of the datasets: See section 3.1.1 about data location. Enter the data location on the local computer or network file system. It should contain all the test images and WSI overview images. This field is required (Figure 13).

The figure shows a software panel titled "Submission_Info". It contains five input fields arranged in two rows. The first row has four fields: "K-number" with the value "K345678", "Sponsor" with "MyCompany", "Device" with "MyDevice", and "FileFormat" with "MyFormat". The second row has a single wide field labeled "Root folder" with the value "N:\demo\input".

Figure 13: Enter the submission information.

4.2 Panel 2: Enter Display Information

The information entered for the display is used to calculate the cropping ratio, which is the ratio of the pixel count of a test image to the pixel count of the screen (Figure 14).

- Display_Model: For information only.
- Width and Height: The pixel count of the screen (Width times Height) will be used as the denominator.

The figure shows a software panel titled "Display_Info". It contains three input fields. The first is "Display_Model" with the value "MyDisplay". The second and third are "Width" with the value "1920" and "Height" with the value "1280".

Figure 14: Enter display information.

4.3 Panel 3: Enter Dataset Information

The information entered in this panel describes the 4 parameters of the datasets: slide, ROI, magnification level, and device (Figure 15).

4.3.1 Enter Slide Names

The user provides the slide names in the Slide_Name table.

- **slide_n**: The number of slides in the dataset (e.g., 8). This variable determines the length (rows) of the Slide_Name table.
- **Slide_Name**: The name of each slide. There are 3 ways of entering the information:
 - a. Double-click on the table and enter the information manually. For example, enter the slide names: 017, 036, 072, 074, 096, 108, 128, and 131.
 - b. In the “Using” field, select “Formula” and enter the formula in the “Formula_Slide” field as a string. Click on the “Gen_Slide” button. The app will use the MATLAB “sprintf” function with the entered formula string to populate the table. The formula should include the variable “i” which iterates from 1 to “slide_n”.
 - c. Prepare the slide names in another document and copy the text into the clipboard. Then in the “Using” field, select “Paste” to paste the text into the table.
- This table will provide a string variable “s”, which represents the *i*-th slide name.

Slide# i = 1 : slide_n Using Formula

Formula_Slide

	Slide_Name (s)
1	017
2	036
3	072
4	074

	Folder_Slide_Name (ss)
1	
2	
3	

Figure 15: Enter slide information.

4.3.2 Enter ROI Names

The user provides the ROI names in the “ROI_Name” table (Figure 16).

- **ROI_n**: The number of the ROIs in the dataset (e.g., 3). This variable determines the length (rows) of the ROI_Name table.
- **ROI_Name**: The name of each ROI. There are 2 ways of entering the information:
 - a. Double-click on the table and enter the information manually. For example, enter the ROI names: roi1, roi2, and roi3.

- b. Enter the formula in the “Formula_ROI” field as a string. Click on the “Gen_ROI” button. The app will use the MATLAB “sprintf” function with the entered formula string to populate the table. The formula should include the variable “j” which iterates from 1 to “ROI_n”.

This table will provide a string variable “r”, which represents the j -th ROI name.

ROI# j = 1 : 3 roi_n

Formula_ROI 'ROI%d'j

Gen_ROI

	ROI_Name (r)
1	roi1
2	roi2
3	roi3

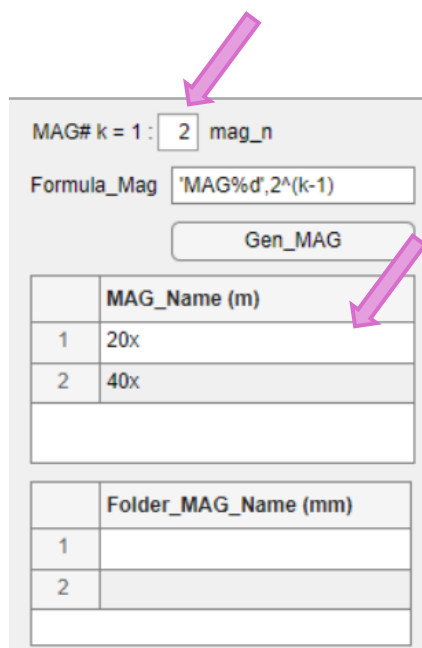
	Folder_ROI_Name (rr)
1	
2	
3	

Figure 16: Enter the ROI information.

4.3.3 Enter Magnification Level Names

The user provides the magnification level names in the “MAG_Name” table (Figure 17).

- mag_n: The number of the magnification levels in the dataset (e.g., 2). This variable determines the length (rows) of the MAG_Name table.
- MAG_Name: The name of each magnification level. There are 2 ways of entering the information:
 - a. Double-click on the table and enter the information manually. For example, enter the magnification level names: 20x and 40x.
 - b. Enter the formula in the “Formula_Mag” field as a string. Click on the “Gen_MAG” button. The app will use the MATLAB “sprintf” function with the entered formula string to populate the table. The formula should include the variable “k” which iterates from 1 to “mag_n”.
- This table will provide a string variable “m”, which represents the k -th magnification level name.



MAG# k = 1 : mag_n

Formula_Mag

	MAG_Name (m)
1	20x
2	40x

	Folder_MAG_Name (mm)
1	
2	

Figure 17: Enter the magnification level information.

4.3.4 Enter Device/Browser Information

The user needs to provide the information about the devices, browsers, or any other variables to be compared in the study. For example, the study may compare two devices (predicate vs. subject), two browsers for the same device (Chrome vs. Firefox), two file formats, etc. All these variables can be entered and considered as different “devices” (Figure 18).

The user provides the device names in the “Device_Name” table.

- device_n: The number of different devices tested in the dataset (e.g., 2). This variable determines the length (rows) of the Device_Name table.
- MAG_Name: The name of each device. There are 2 ways of entering the information:
 - a. Double-click on the table and enter the information manually. For example, enter the two device names: “predicate” and “subject”.
 - b. Enter the formula in the “Formula_Device” field as a string. Click on the “Gen_Device” button. The app will use the MATLAB “sprintf” function with the entered formula string to populate the table. The formula should include the variable “w” which iterates from 1 to “device_n”.
- This table will provide a string variable “d”, which represents the *w*-th device name.

Device# w = 1 : 2 device_n

Formula_Device 'device%1d',w

Gen_Device

	Device_Name (d)
1	predicate
2	subject

	Folder_Device_Name (dd)
1	
2	

Figure 18: Enter the device/browser information.

4.4 Panel 4: Construct Path Name

In this panel, the user enters two formulas to construct the path name (under the root folder) of each test image using the 4 parameters established in panel #3. The path name is constructed by two strings:

<Folder_Naming>/<File_Naming>

The app will use the MATLAB “sprintf” function with the two strings entered in the Folder_Naming and File_Naming fields to populate the table. The user can use any of the variables (l, j, k, w, s, r, m, d) in the formulas. Enter an empty string (”) in the Folder_Naming field if the folder name is not used.

Figure 19 shows how to use the formulas to construct the path names for the demo datasets.

Screenshot_Filename

Folder_Naming '%s',d' File_Naming '%s_%s_%s.png',s,r,m' Gen_Pathname 96/96 (100%)

	Sli...	ROI#	M...	De...	Pathname	Found
1	1	1	1	1	predicate/017_roi1_20x.png	☒
2	1	1	1	2	subject/017_roi1_20x.png	☒
3	1	1	2	1	predicate/017_roi1_40x.png	☒
4	1	1	2	2	subject/017_roi1_40x.png	☒
5	1	2	1	1	predicate/017_roi2_20x.png	☒

Figure 19: Populate and check all file names.

Once the two formulas are entered, click on the “Gen_Pathname” button to populate the table (Figure 20). The app will check whether each file exists and mark the result in the “Found” column. The app displays the inventory result, e.g., '96/96 (100%)'. If the inventory result is not 100%, scroll the table to locate the path name that was not found. Check the path name to verify whether the image file exists and whether the formulas were entered correctly.

⚠ IMPORTANT: All files must be present (100%) before proceeding with the following ΔE analysis. If files are missing, review the dataset organization and naming convention.

📌 NOTE: If the color difference analysis must be conducted with the partial datasets, the user can exclude the missing files by reducing the data range (e.g., changing from 8 slides to 7 slides).

The screenshot shows the 'Screenshot_Filename' panel. It includes input fields for 'Folder_Naming' (containing '%s',d) and 'File_Naming' (containing '%s_%s_%s.png',s,r,m). A 'Gen_Pathname' button is present, and below it, the text '96/96 (100%)' indicates the completion status. A table lists five rows of file paths, each with a 'Found' checkbox that is checked. Pink arrows point to the 'Gen_Pathname' button, the 'File_Naming' field, and the 'Found' column.

	Sli...	ROI#	M...	De...	Pathname	Found
1	1	1	1	1	predicate/017_roi1_20x.png	☒
2	1	1	1	2	subject/017_roi1_20x.png	☒
3	1	1	2	1	predicate/017_roi1_40x.png	☒
4	1	1	2	2	subject/017_roi1_40x.png	☒
5	1	2	1	1	predicate/017_roi2_20x.png	☒

Figure 20: Check the inventory result, populated path names, and missing files.

4.5 Panel 5: Enter WSI Overview Filenames

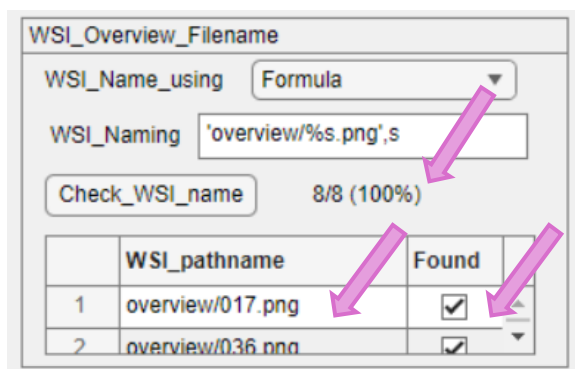
In this panel, the user enters the path name (under the root folder) of each WSI overview file. The filenames can be entered using a formula based on the slide parameters (i and s) established in panel #3, the copy-and-paste method, or direct editing (Figure 21).

The screenshot shows the 'WSI_Overview_Filename' panel. It includes a 'WSI_Name_using' dropdown set to 'Formula', a 'WSI_Naming' field containing 'overview/%s.png',s', and a 'Check_WSI_name' button. Below the button, the text '8/8 (100%)' indicates the completion status. A table lists two rows of file paths, each with a 'Found' checkbox that is checked. Pink arrows point to the 'WSI_Name_using' dropdown, the 'WSI_Naming' field, the 'Check_WSI_name' button, and the 'WSI_pathname' column.

	WSI_pathname	Found
1	overview/017.png	☒
2	overview/036.png	☒

Figure 21: Enter the WSI overview image information.

When the “Check_WSI_name” button is clicked, the app will check whether each file exists, mark the “Found” column, and displays the inventory result (Figure 22).



WSI_Overview_Filename

WSI_Name_using Formula

WSI_Naming 'overview/%s.png',s

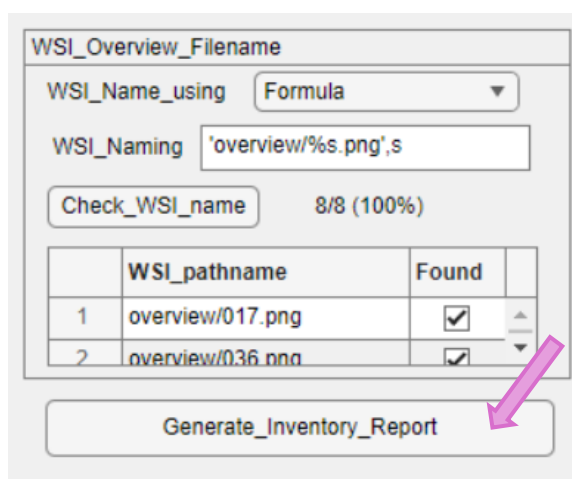
Check_WSI_name 8/8 (100%)

	WSI_pathname	Found
1	overview/017.png	☒
2	overview/036.png	☒

Figure 22: Populate and check the WSI overview images.

4.6 Panel 6: Generate Data Inventory Report (Optional)

The user can generate an inventory report by clicking on the 'Generate_Inventory_Report' button (Figure 23). A Word document will be generated automatically. It will be named based on the K-number (e.g., K345678_inventory.docx).



WSI_Overview_Filename

WSI_Name_using Formula

WSI_Naming 'overview/%s.png',s

Check_WSI_name 8/8 (100%)

	WSI_pathname	Found
1	overview/017.png	☒
2	overview/036.png	☒

Generate_Inventory_Report

Figure 23: Generate inventory report.

4.7 Panel 7: Run the ΔE_{00} Calculation

If the datasets pass the inventory check, the user can proceed to the next 3 panels on the right half of the GUI to conduct the pixelwise comparison.

First, the user clicks on the "Create_RTA" button to confirm the information entered in the previous 6 panels and start the pixelwise comparison (Figure 24).

RTA_Analysis

Create_RTA Run_All

Check_Completeness

Gen_ROI_chart

Comp# x = 1 : 1 comp_n

	Predicate	Subject
1	1	2

Calculate_dE

Figure 24: Start the pixelwise comparison session.

Next, the user needs to specify the paired datasets to be compared. The “comp_n” variable indicates how many comparisons will be conducted (e.g., 1) and determines the length (number of rows) of the comparison table. In each row, the user chooses two entries from the “Device_Name” table and enter their indices. For the example shown in Figure 25, only one comparison will be conducted. Row #1 indicates the first device “predicate” (index=1) and the second device “subject” (index=2) will be compared.

Device# w = 1 : 2 device_n*

Formula_Device 'device%1d',w

Gen_Device

*	Device_Name (d)
1	
2	

	Folder_Device_Name (dd)
1	predicate
2	subject

Comp# x = 1 : 1 comp_n*

*	Predicate	Subject
1	1	2

Calculate_dE

Store_Load_Form

Output_filename tock345678_JPG.mat Save

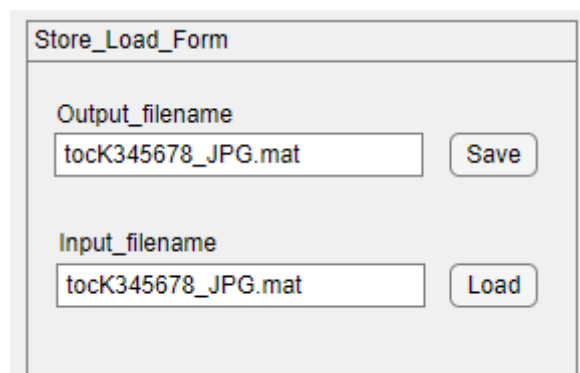
Input_filename tock345678_JPG.mat Load

Figure 25: Specify the paired datasets to compare.

Finally, click on the 'Run_All' button (Figure 24) to start the calculation and wait for processing to complete (progress will be displayed).

4.8 Panel 8: Store/Load the Variables

In this panel, the user can store the data entered in the GUI for future use, by entering the “Output_filename” (e.g., tock345678_JPG.mat), following with clicking “Save” button (Figure 26). Similarly, the user can click the Load button to load an existing variable mat file (e.g., tock345678_JPG.mat).

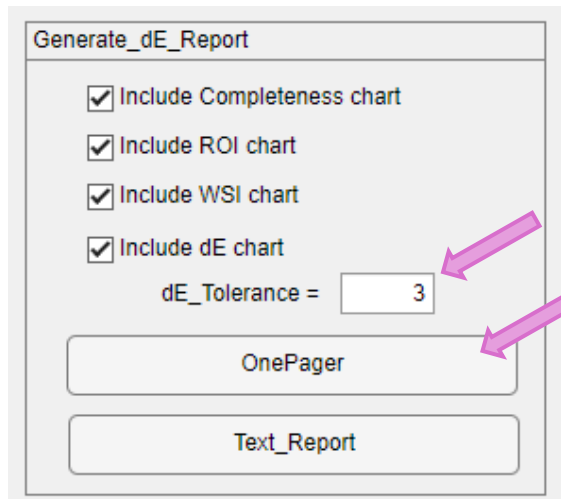


The image shows a dialog box titled "Store_Load_Form". It contains two sections. The first section is labeled "Output_filename" and has a text input field containing "tock345678_JPG.mat" and a "Save" button to its right. The second section is labeled "Input_filename" and has a text input field containing "tock345678_JPG.mat" and a "Load" button to its right.

Figure 26: Save information entered in the GUI for future use.

4.9 Panel 9: Enter Tolerance Threshold Value for ΔE_{00} and Generate One-Pager Report

In this panel, enter the tolerance threshold value for ΔE_{00} (e.g., 3 ΔE_{00}) and select whether to include the completeness chart, ROI chart, and WSI chart in the report. Then click on the 'OnePager' button (Figure 27).



The image shows a dialog box titled "Generate_dE_Report". It contains four checked checkboxes: "Include Completeness chart", "Include ROI chart", "Include WSI chart", and "Include dE chart". Below these is a label "dE_Tolerance =" followed by a text input field containing the value "3". Two pink arrows point to the input field and the "OnePager" button. At the bottom are two buttons: "OnePager" and "Text_Report".

Figure 27: Enter the tolerance threshold and generate the report.

The one-page report will be generated on the screen (Figure 28).

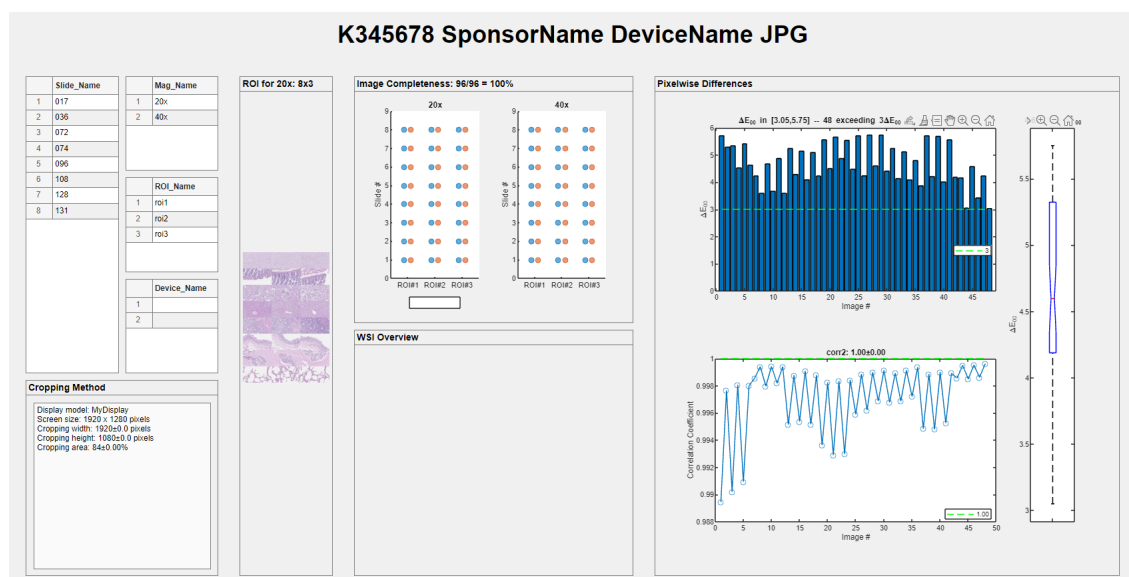


Figure 28: One-pager report.

5 Interpret the Report

The one-pager report provides a dashboard-style, comprehensive summary of the pixelwise comparison study (Figure 29). This section explains each component of the report generated by IVIES.

The report contains six panels:

- Input Data Range
- Cropping Area Ratio Check
- ROI Content Check
- Data Completeness Check
- WSI Overview Completeness Check
- Color Difference Analysis

5.1 Panel 1: Input Data Range

	Slide_Name
1	017
2	036
3	072
4	074
5	096
6	108
7	128
8	131

	Mag_Name
1	20x
2	40x

	ROI_Name
1	roi1
2	roi2
3	roi3

	Device_Name
1	
2	

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5.2 Panel 2: Cropping Area Ratio Check

This panel shows the pixel counts of the display and the cropped images (Figure 31).

- **Display Size:** The pixel count of the display used in the study (e.g., 3840×2160 pixels)
- **Cropping Size:** The dimensions of the ROI images (e.g., 1920×1280 pixels)
- **Ratio:** Proportion of the display area used (e.g., 30%)

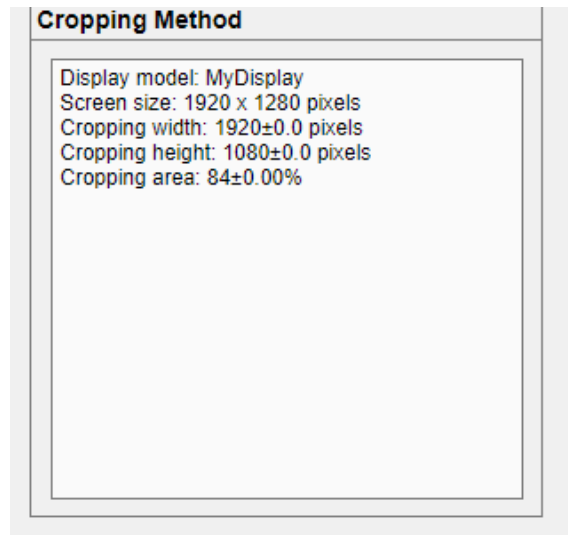


Figure 31: Cropping area ratio check.

5.3 Panel 3: ROI Content Check

This panel displays the thumbnails of the test images in a grid:

- **Rows:** 8 slides
- **Columns:** 3 ROIs per slide

By visual inspection of the example shown in Figure 32, the user can confirm:

- All images are H&E stained
- Different features were selected in the ROIs from the same slide
- No blank, uniform, or repeated ROIs present

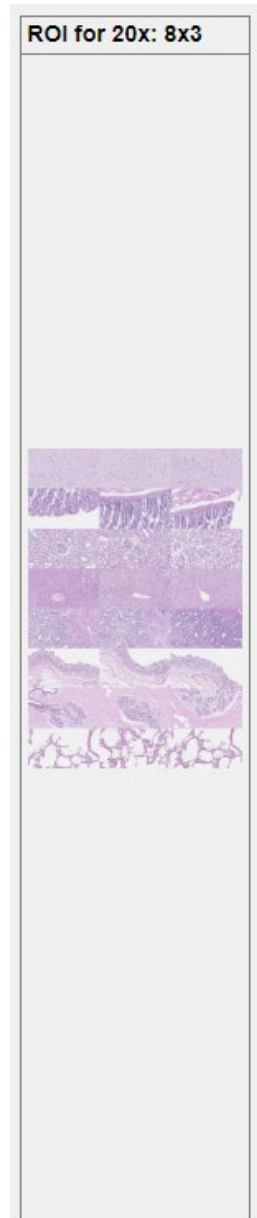


Figure 32: ROI content check.

5.4 Panel 4: Data Completeness Check

This panel visualizes the completeness of the datasets (Figure 33).

- **Blue markers:** Predicate device images
- **Red markers:** Subject device images
- **Left bank:** 20x magnification
- **Right bank:** 40x magnification

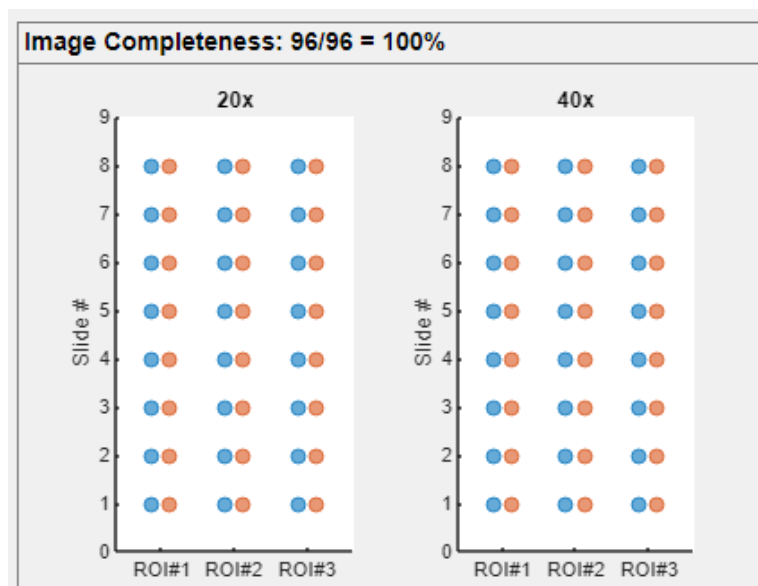


Figure 33: Data completeness check.

5.5 Panel 5: WSI Overview Image Completeness Check

This panel displays thumbnail views of the WSI overview images (Figure 34).



Figure 34: WSI overview completeness check.

5.6 Panel 6: Color Difference Analysis

This panel shows the primary results of color difference analysis (Figure 35). The bar chart shows the 95th percentile ΔE_{00} values of the 48 comparisons (8 slides $\times$ 3 ROIs $\times$ 2 magnifications). The 95th percentile ΔE_{00} values range between 3.05 and 7.75 ΔE_{00} . When the tolerance threshold is 3 ΔE_{00} , 48 out of 48 comparisons exceed the tolerance threshold. The line plot shows the correlation coefficients of the 48 comparisons based on the MATLAB “corr2” function. The boxplot on the right shows the distribution of the 95th percentile ΔE_{00} values where the mean is 4.67 ΔE_{00} .

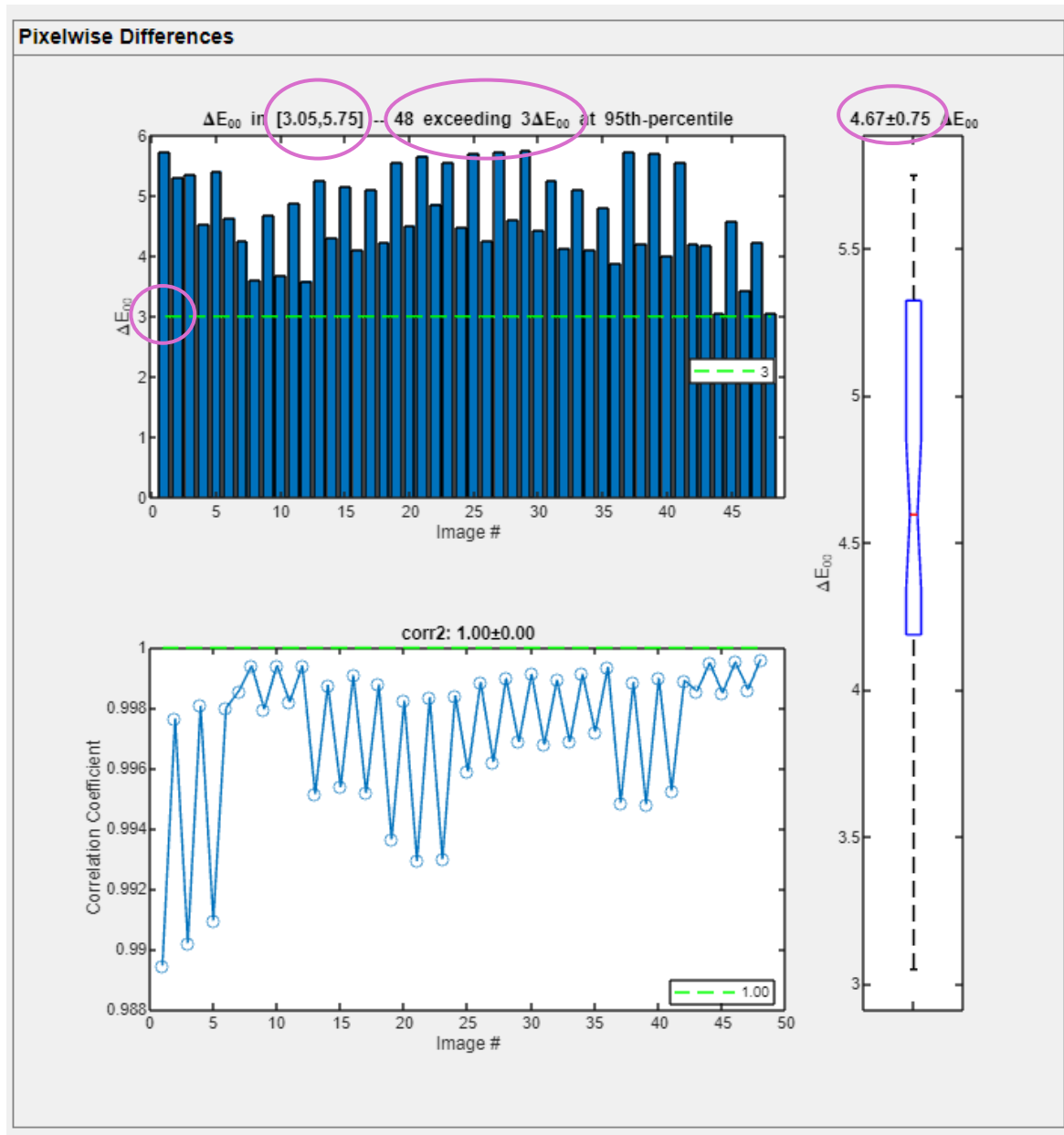


Figure 35: Color difference analysis.

5.7 Quality Control for Input Data

Quality control of input data is essential for obtaining valid and reliable results from pixelwise comparison studies. Figure 36 shows how the report assists in quality control to comply with the FDA guidelines.

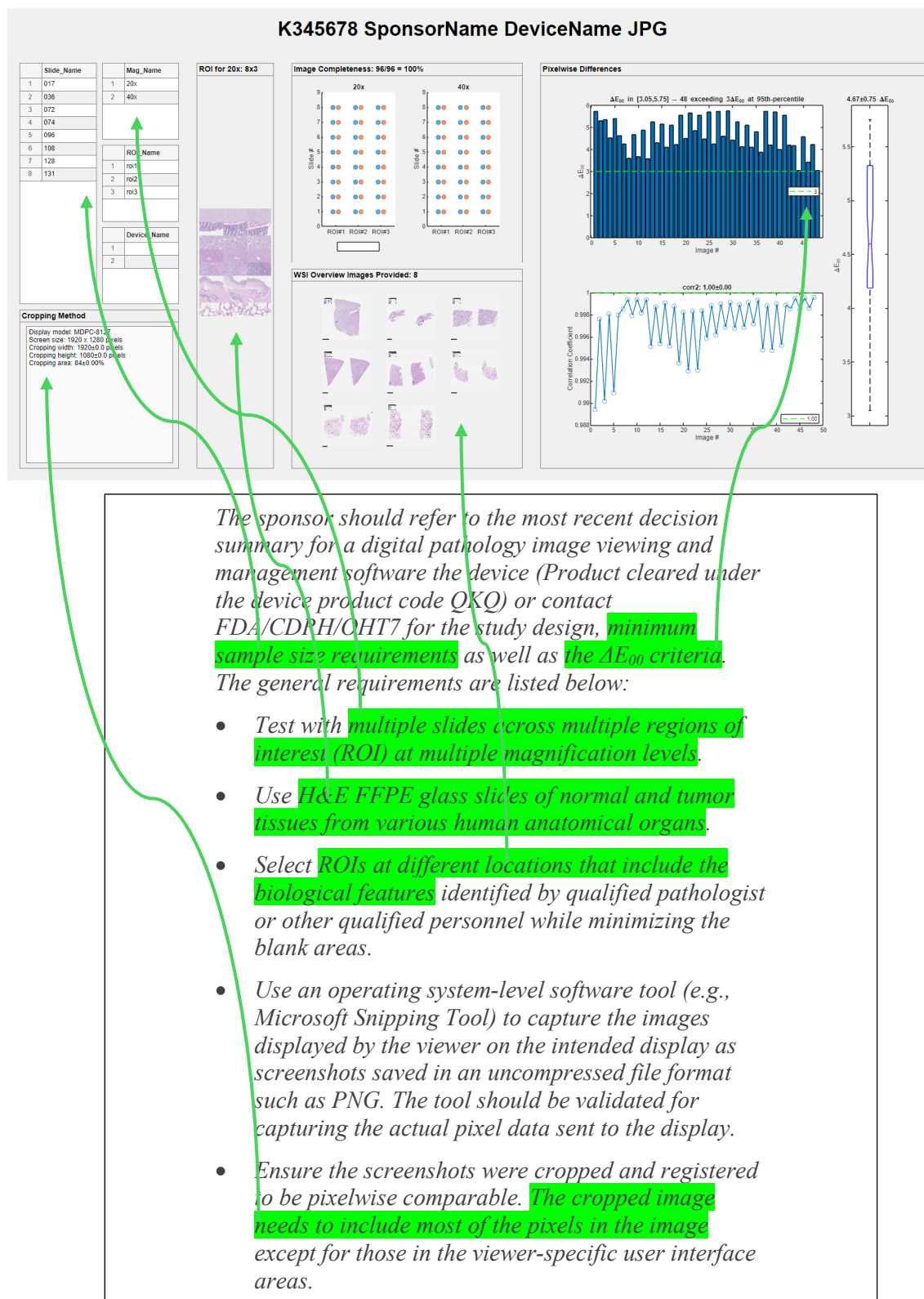


Figure 36: Quality control.

6 Contact Information

Questions or comments concerning this tool may be directed to:

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Appendix A: FDA's guidelines for file naming convention

1. For each supported scanner, the test images should include all combinations of the following four parameters:
 - i. Whole-slide images scanned from different H&E-stained glass slides (e.g., slide#1~slide#30)
 - ii. Regions of interest (ROIs), that are of adequate size and from different locations in the tissue in each scanned whole-slide image (e.g., ROI#1, ROI#2, and ROI#3). Select ROIs that include the biological features identified by qualified pathologist or other qualified personnel while minimizing the blank areas.
 - iii. 2 (or more) Magnification levels for each ROI depending on the scanner (e.g., 20x and 40x)
 - iv. All supported device-browser combinations for each zoom level depending on your intended use (e.g., predicate-chrome and subject-chrome)

For example, if your viewer is a standalone application or a web-based application that supports only one 'browser' (e.g., Chrome or Edge), and you choose two zoom levels, 30 glass slides and 3 ROIs per each glass slide in the testing, then you should provide a total of **30 slides x 3 ROIs x 2 zoom levels x 2 devices** images. Please be advised when you conduct the test, all pixels in each screenshot should be evaluated pixelwise except for those in the viewer-specific user interface areas. In addition, image-pairs should be registered and ready to be compared pixelwise.

For ease of review, provide a summary table as shown below.

Testing Data	Number	Example
Slides	30	slide#1 ~ slide#30
ROI per slide	3	ROI#1, ROI#2, and ROI#3
Zoom Levels per ROI	2	20x, 40x
Browsers	1	Chrome
Total images pairs	180	30 x 3 x 2 x 1 in example

2. For the test image files, you can use your own naming convention and your own folder structure. However, please provide the mapping such that each image can be identified and located with the four numerical parameters – slide, ROI, zoom level, and device-browser. Please refer to below general examples.

- i. Fill out the following table "SLIDE_ID" to number the slide names used in your dataset

SLIDE_ID	Example
1	"Breast"
2	"Colon"
3	"Skin"
...	
30	"Lung"

- ii. Fill out the following table “ROI_ID” to number the ROI names used in your dataset

ROI_ID	Example
1	“ROI_1”
2	“ROI_2”
3	“ROI_3”

- iii. Fill out the following table “MAG_ID” to number the zoom level names used in your dataset

MAG_ID	Example
1	“20x”
2	“40x”

- iv. Fill out the following table “DEVICE_ID” to number the device-browser names used in your dataset

DEVICE_ID	Example
1	“IMS_Chrome”
2	“MyViewer_Edge”

Then provide a function to construct the filename (or pathname) using the four parameters. For example, your naming convention, f , could be:

$$f(\text{SLIDE_ID}, \text{ROI_ID}, \text{MAG_ID}, \text{DEVICE_ID}) = [\text{SLIDE_ID}]/[\text{ROI_ID}]/[\text{DEVICE_ID}]_{[\text{MAG_ID}]}$$

Then we can find the image of slide #2, ROI #3, zoom level #1 from the subject device in the following folder:

$$f(2,3,1,2) = \text{“Colon/ROI_3/MyViewer_Edge_20x.png”}$$

- Please provide overview images of the glass slides with the selected ROIs annotated.
- Please describe the whole-slide images with overview images and a list of descriptions of tissue type and stain as in the following example table.

SLIDE_ID	Stain	Tissue/Tumor Type
1	H&E	Breast
2	H&E	Colon
3		Skin
...		
30	H&E	Lung

- The above data in a specified format is needed for FDA to conduct an adequate review to evaluate the performance of your device and determine substantial equivalence to the predicate which had appropriate presentation of data.